

Math 290

Practice Quiz 2 — Version B Solutions

1(a) Answer: D. To prove “if $2 \mid x$, then x even” contrapositively, you assume x is *not* even and show $2 \nmid x$. The student instead assumed x is even and showed $2 \mid x$, which proves the converse. A correct direct proof: assume $2 \mid x$; then $x = 2c$ for some $c \in \mathbb{Z}$ by the definition of divides, so x is even. (*Based on Exercise 7.4.*)

1(b) Answer: False. The converse of $P \Rightarrow Q$ is $Q \Rightarrow P$: “If x^2 is even, then x is even.” The given sentence is the *inverse*. (*Based on Section 7.C.*)

1(c) Answer: True. The premise $x \in \emptyset$ is never true (vacuous), and the conclusion $x > 0$ is always true for $x \in \mathbb{N}$ (trivial). (*Based on Example 6.15(3).*)

2. (*Based on Exercise 8.5.*)

Solution. Let $n \in \mathbb{Z}$.

($\Rightarrow$) We work directly. Assume $3 \mid n$, so $n = 3k$ for some $k \in \mathbb{Z}$. Then $n^2 = 9k^2 = 3(3k^2)$, and $3k^2 \in \mathbb{Z}$, so $3 \mid n^2$.

($\Leftarrow$) We work contrapositively. Assume $3 \nmid n$. Then there are two cases. Case 1: $n = 3k + 1$ for some $k \in \mathbb{Z}$. Then $n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$. Case 2: $n = 3k + 2$ for some $k \in \mathbb{Z}$. Then $n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$. In both cases $n^2 \equiv 1 \pmod{3}$. Since every integer is of exactly one of the forms $3m$, $3m + 1$, $3m + 2$, n^2 is not of the form $3m$, so $3 \nmid n^2$. $\square$

3. (*Based on Exercise 9.9.*)

Solution. Let $x, y \in \mathbb{Z}$. Assume, by way of contradiction, that $33x + 132y = 57$. Then

$$57 = 33x + 132y = 11(3x + 12y),$$

and $3x + 12y \in \mathbb{Z}$, so $11 \mid 57$. But $57 = 11 \cdot 5 + 2$ lies strictly between $11 \cdot 5 = 55$ and $11 \cdot 6 = 66$, so $57/11$ is not an integer and $11 \nmid 57$. This is a contradiction, so $33x + 132y \neq 57$. $\square$

(Equally good: factor out 33 and note $33 \nmid 57$.)

4. (*Based on Section 10 — the $\cup$ version of the posted quiz’s problem 4.*)

Solution. Let A and B be sets.

($\Leftarrow$) Assume $A \subseteq B$. First we show $A \cup B \subseteq B$. Let $x \in A \cup B$. Then $x \in A$ or $x \in B$ (definition of union). Case 1: $x \in A$. Then $x \in B$ (definition of subset, since $A \subseteq B$). Case 2: $x \in B$. Then $x \in B$ (assumption of Case 2). In both cases $x \in B$ (proof by cases), so $A \cup B \subseteq B$ (definition of subset). Next we show $B \subseteq A \cup B$. Let $x \in B$. Then $x \in A$ or $x \in B$ (tautology $Q \Rightarrow P \vee Q$), so $x \in A \cup B$ (definition of union). Thus $B \subseteq A \cup B$ (definition of subset). Hence $A \cup B = B$ (definition of set equality).

($\Rightarrow$) Assume $A \cup B = B$. Let $x \in A$. Then $x \in A$ or $x \in B$ (tautology $P \Rightarrow P \vee Q$), so $x \in A \cup B$ (definition of union). Since $A \cup B = B$, we get $x \in B$ (substitution, using the assumption $A \cup B = B$). Therefore $A \subseteq B$ (definition of subset). $\square$